

Mission-critical communications for public safety and emergency response using 4G/5G network slicing for reliability.

Public / Government sector × Tactical and mission-critical communications × Network slicing · OS144 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
61 is the world moving	51 can we play, can we win	MEDIUM 0 named in the evidence	€220m observed evidence · per year	LATER research and standards activi...	L1 14 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

This opportunity targets public safety and emergency response agencies that need reliable, mission-critical communications. By leveraging 4G/5G network slicing, Orange can deliver dedicated, secure communication channels that meet strict availability and latency requirements. The need is urgent as agencies modernize their communication infrastructure to support advanced applications like MCVideo and MCDData.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Observed: tendered contract value, annualised	€220m €220m – €482m	€220m €220m – €482m	€2.6m €2.6m – €5.8m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

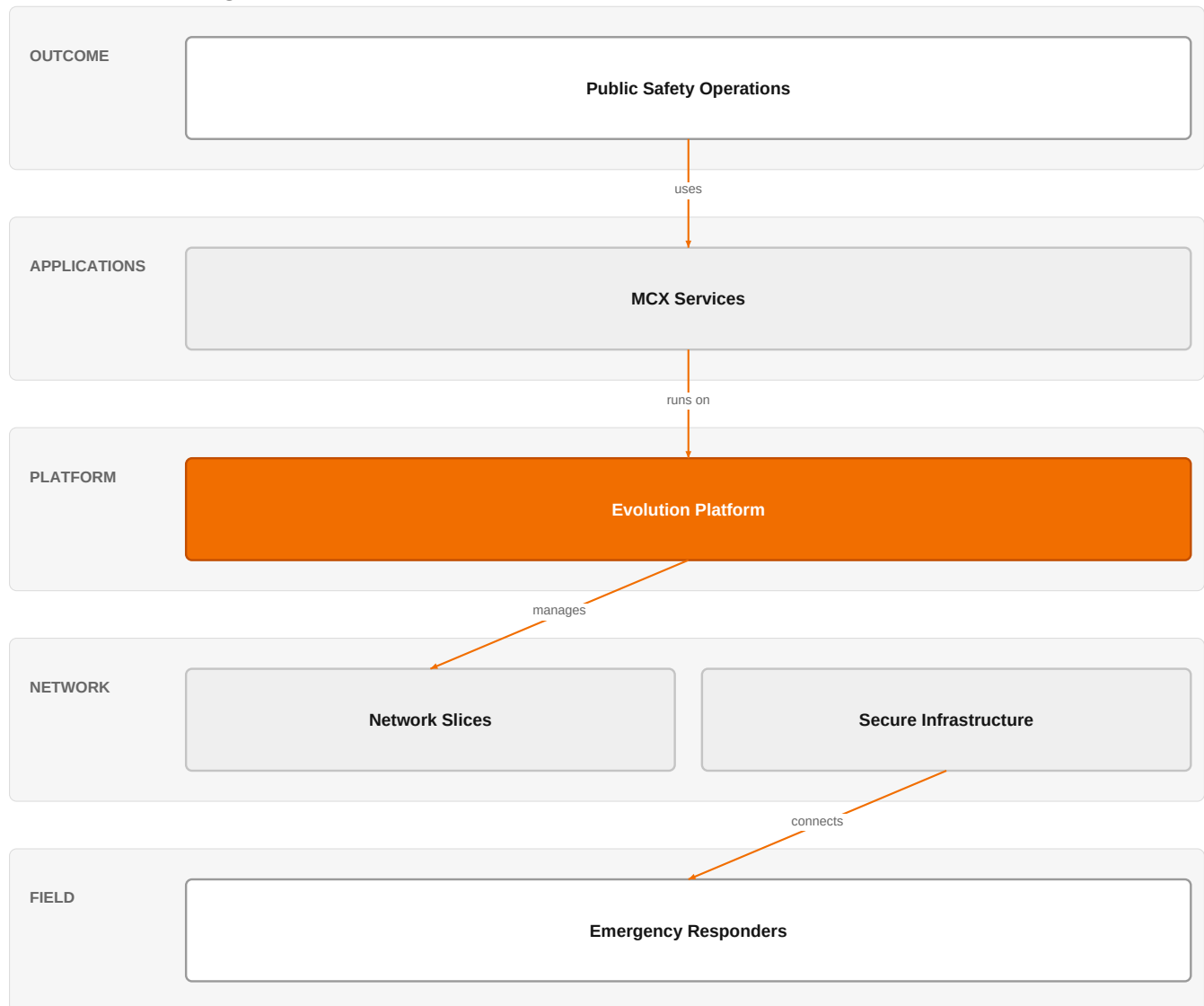
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€220m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 34 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

The solution, in one picture

Mission-Critical Slicing Solution



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This diagram shows how Orange's Evolution Platform orchestrates secure network slices to deliver mission-critical services to emergency responders.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Mission-critical services (MCX) are standardized, including MCPTT, MCVideo, and MCDData, with stringent requirements for availability, reliability, latency, and security, essential for public safety and emergency response. Network slicing enables multiple virtualized network instances over shared infrastructure, each tailored to specific QoS profiles, making it a key enabler for URLLC services. However, network slicing introduces security vulnerabilities due to its dynamic virtualization layer, requiring robust security measures. SLAs, including security SLAs, are crucial for defining QoS expectations and accountability, which is critical for mission-critical communications.

Sources: SIG-C0BAE4D32A34 openalex.org 2025-08-19; SIG-FD57D57901DD Elsevier BV 2026-01-01; SIG-263E4357EB27 openalex.org 2025-11-08; SIG-AEA38615C065 openalex.org 2026-03-26

Who buys, and why now

The buyer is typically the chief technology officer or head of communications at a public safety agency (e.g., police, fire, emergency medical services) or a government department responsible for emergency response. The pain is operational: current communication systems lack the reliability, security, and bandwidth to support modern mission-critical applications, especially during large-scale incidents. The trigger is often a major incident that exposed communication failures, or a mandate to upgrade to broadband-based MCX services.

Sources: SIG-C0BAE4D32A34 openalex.org 2025-08-19

Why it is hot now — every claim bound to a dated source

- Mission-critical services are essential for public safety and emergency response, where communication performance directly affects operational effectiveness. [SIG-C0BAE4D32A34]
- SLA defines QoS expectations for service delivery, which is critical for mission-critical communications. [SIG-AEA38615C065]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a mission-critical communication solution using its Evolution Platform as the core, integrating network slicing capabilities. The engagement would start with a needs assessment using Orange's IoT and Digital experts, followed by a design phase leveraging Orange Group researchers for optimization. The solution would be deployed on Orange's secure infrastructure, certified with ISO 27001, SecNumCloud, and CISPE, ensuring sovereignty and trust. A proof-of-concept could be based on the Grand Paris Sud reference, demonstrating real-world applicability.

Why Orange can win

Orange can win because it combines deep expertise in defense and security with certified infrastructure (ISO 27001, SecNumCloud, CISPE) that meets the stringent security requirements of public safety. The Evolution Platform provides a flexible foundation for network slicing, and the Grand Paris Sud reference demonstrates successful deployment in a complex environment. However, the assets are thin; Orange must leverage its research capabilities to differentiate on optimization and security.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Evolution Platform	offer	L1	offer provides the technology in a matching domain	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
SecNumCloud	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
CISPE	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Digital experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Grand Paris Sud	reference	SUP	published reference in this vertical, different use case	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 6.6; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitor	Category	Position here	Basis	Evidence
Colt Technology Services	Telecom operator peer	sells Network slicing; competes in Connectivity Solutions	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	sells Network slicing; active in Public / Government sector; competes in Connectivity Solutions	structural	—
Vodafone Business	Telecom operator peer	sells Network slicing; competes in Connectivity Solutions	structural	—
Boldyn Networks	Network equipment vendor	sells Network slicing; active in Public / Government sector; competes in Connectivity Solutions	structural	—

Ericsson	Network equipment vendor	sells Network slicing; competes in Connectivity Solutions	structural	—
Huawei	Network equipment vendor	sells Network slicing; competes in Connectivity Solutions — also an Orange partner	structural	—
Nokia	Network equipment vendor	sells Network slicing; competes in Connectivity Solutions	structural	—
Accenture	Global systems integrator	active in Public / Government sector	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 What is your current communication system for mission-critical operations, and what are its biggest limitations during large-scale incidents?
- 2 Have you experienced any communication failures in recent emergencies that impacted response effectiveness?
- 3 Are you planning to adopt MCX services like MCPTT, MCVideo, or MCDData in the next 12-24 months?
- 4 What are your specific requirements for network reliability, latency, and security in mission-critical scenarios?
- 5 Do you have any existing SLAs with your current communication provider, and how are they enforced?
- 6 How do you currently ensure the security of your communication networks, and what compliance standards do you need to meet?

Objections you will meet

They will say	You can answer
Network slicing is not mature enough for mission-critical use.	You're right that commercial deployments are still emerging, but the technology is standardized and proven in trials. Orange can offer a proof-of-concept to demonstrate reliability and security in your environment.
We already have a reliable communication system; why change?	Your current system may be reliable for voice, but it likely lacks the bandwidth and flexibility for video and data applications that are becoming essential. Network slicing can provide dedicated, secure channels for these new services without compromising reliability.
We are concerned about security and sovereignty.	Security is a top priority. Orange's infrastructure is certified to the highest standards (ISO 27001, SecNumCloud, CISPE), and we have deep expertise in defense and security. We can provide a sovereign solution that meets your compliance needs.

Risks and unknowns

The main risk is that network slicing for mission-critical communications is still in research and standards phase; commercial deployments are limited. Security vulnerabilities in slicing are not fully understood, and SLAs for slicing are not mature. Additionally, the customer may be hesitant to rely on a single operator for such critical services, and regulatory approval may be required.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a public-safety tender, using the Evolution Platform and referencing Grand Paris Sud as a proof point, and outline how ISO 27001, SecNumCloud, and CISPE certifications support the security and compliance requirements.
Sales	In a conversation with a European public-safety agency's head of communications, say: 'We have a platform, Evolution Platform, that can support mission-critical communications with network slicing, and we're exploring how to meet your reliability needs.'
Strategist / Innovator	Deep-dive brief on the Evolution Platform's network slicing capabilities, focusing on how they map to mission-critical QoS requirements, and commission a study on the gap between current 4G/5G slicing standards and the SLA expectations of public safety agencies.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-D8CAA774FDC4	2026-07-01	Next Research	1	Securing 5G network slicing for the smart grid network
SIG-AEA38615C065	2026-03-26	openalex.org	1	Service Level Agreement (SLA) and Security SLA (SecSLA): A Comprehensive Survey
SIG-C1DEE33668FF	2026-03-02	2026 21st Wireless On-Dem...	1	Proactive Resource Optimization for Heterogeneous 5G Network Slicing
SIG-FD57D57901DD	2026-01-01	Elsevier BV	1	Multi-Objective Optimization of Latency, Reliability, and Throughput in 5G Network Slicing for ...
SIG-ECAF07DC10C9	2025-12-18	openalex.org	1	From 5G to 6G: A Survey on Security, Privacy, and Standardization Pathways
SIG-88237402830B	2025-11-28	SN Computer Science	1	Optimizing 5G Network Slicing for Heterogeneous Services
SIG-A5AB1DD30A6D	2025-11-10	openalex.org	1	E2E Network Slicing for Enhanced Cybersecurity, Orchestration, Automation and Response in 5G/6G...
SIG-263E4357EB27	2025-11-08	openalex.org	1	Investigating 5G Network Slicing Security Vulnerabilities Using Artificial Intelligence–Driven ...
SIG-EBB2157E5B89	2025-11-01	Acta Scientific Medical S...	1	Resilient AIDriven Network Slicing for DisasterAware 6G Infrastructures
SIG-2AE1E14EE912	2025-10-27	Institute of Electrical a...	1	ARCANE: Adversarial Resilience and Adaptive Network Slicing for UAV-based MEC
SIG-D4CA61FE83AD	2025-08-29	openalex.org	1	AI-powered cyber resilience: a reinforcement learning approach for automated threat hunting in ...
SIG-A311539B246C	2025-08-22	2025 2nd International Co...	1	An Intelligent Framework for Network Slicing to Enhancing Resource Scheduling in 5G Network
SIG-C0BAE4D32A34	2025-08-19	openalex.org	1	Mission-Critical Services in 4G/5G and Beyond: Standardization, Key Challenges, and Future Pers...
SIG-67C4E4DA4852	2026-07-01	arxiv.org	3	Latency-Sensitive 5G RAN Slicing for Industry 4.0

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding); diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:16+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:19+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:57:27+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.